

# Pearls AQI Predictor: Interactive Forecasting Dashboard and Public Deployment

Milestone 7 Technical Report

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**Abstract**—This report describes the development and public deployment of the user-facing dashboard for the Pearls AQI Predictor. The dashboard, built with Streamlit, loads the trained model and recent features from the Hopsworks Feature Store and Model Registry, computes a three-day-ahead Air Quality Index (AQI) forecast for Karachi using recursive multi-step prediction, and presents the results through an interactive interface. It displays the current air quality, a colour-coded three-day forecast, an interactive trend chart, and an automated alert for hazardous AQI levels. The application was deployed publicly on Streamlit Community Cloud, completing the serverless end-to-end system.

**Index Terms**—Streamlit, dashboard, data visualization, recursive forecasting, deployment, AQI

## I. INTRODUCTION

The final user-facing component of the Pearls AQI Predictor is a web dashboard that makes the model’s forecasts accessible to the public. According to the project requirements, the application must load the model and features from the Feature Store, compute real-time predictions for the next three days, present them on a simple and descriptive dashboard, and include alerts for hazardous AQI levels. This milestone delivers that dashboard and deploys it publicly.

Streamlit was selected for the web application because it allows interactive data applications to be built entirely in Python, integrates cleanly with the existing project code, and can be deployed for free on Streamlit Community Cloud, preserving the serverless nature of the system.

## II. DASHBOARD ARCHITECTURE

When the dashboard loads, it authenticates with Hopsworks and retrieves two artifacts: the best trained model (a Random Forest) from the Model Registry, together with its feature scaler and feature list, and the most recent engineered features from the Feature Store. To avoid repeated downloads, these resources are cached for the duration of the session.

The current AQI is read from the most recent record and displayed prominently with its health category. The interface is organized into four sections: the current air quality, the three-day forecast, a health alert, and a combined trend-and-forecast chart.

## III. THREE-DAY FORECAST METHOD

The trained model predicts AQI 24 hours ahead. To produce a three-day forecast, recursive multi-step prediction is used: the model first predicts the AQI for the next day; this prediction is then appended to the recent history as if it were an observed value, and the model predicts the following day; the process repeats for the third day.

This approach directly satisfies the requirement to compute predictions for the next three days. Its principal limitation is that prediction errors accumulate across steps, so the forecast for later days is less reliable than for the first day. This limitation is disclosed directly on the dashboard, ensuring the forecast is presented honestly.

## IV. VISUALIZATION AND ALERTS

The three-day forecast is presented as a set of colour-coded cards, where the colour corresponds to the standard US EPA AQI health categories (for example, green for good, orange for unhealthy for sensitive groups, and red for unhealthy). An interactive Plotly chart shows the recent measured AQI trend alongside the forecast, allowing the user to visually relate the prediction to recent conditions.

To satisfy the requirement for hazardous-level alerts, the dashboard evaluates the maximum forecast AQI. If any of the three forecast days exceeds the unhealthy threshold, a prominent warning is displayed identifying the affected day and advising sensitive groups to limit outdoor activity. Otherwise, a reassuring message confirms that no hazardous levels are expected.

## V. PUBLIC DEPLOYMENT

The dashboard was deployed on Streamlit Community Cloud, linked directly to the project’s GitHub repository. The Hopsworks API key was supplied through Streamlit’s encrypted secrets mechanism rather than being embedded in the code, maintaining the same security practice used in the automated pipelines. The deployment is publicly accessible via a shareable URL, allowing anyone to view the live forecast. Because the application reads from the continuously updated Feature Store and Model Registry, it automatically reflects the latest data and model produced by the automated pipelines.

## VI. CONCLUSION

This milestone delivered an interactive, publicly deployed dashboard that completes the end-to-end Pearls AQI Predictor system. The dashboard retrieves the latest model and features from Hopsworks, produces a three-day AQI forecast, visualizes it clearly, and warns users of hazardous air quality. Together with the automated pipelines, it forms a fully functional, serverless air quality forecasting service for Karachi, fulfilling the core objectives of the project.